

- Q.15 Write a brief note on opportunities in by-product utilization in food industry.
- Q.16 Give some examples of by-product derived ingredients.
- Q.17 Write a brief note on functional properties of protein.
- Q.18 Explain the applications of by product derived ingredients.
- Q.19 How waste can be minimized during food processing methods.
- Q.20 What do you understand by eco-efficiency of by-product utilization?
- Q.21 Write a brief note on solvent extraction techniques for oil extraction.
- Q.22 Discuss various success stories in by-products utilization.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x8=16)
- Q.23 Explain various thermal and non thermal methods for food by-products.
- Q.24 Explain the importance of by-product utilization in food industry in detail.
- Q.25 Explain the bio processing of food by products in detail. Give their benefits.

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6th Sem/ Food Technology Subject : Food by-Products Utilization

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 What is the environmental benefit of adopting a circular economy approach to by-product utilization?
- Reduction in waste generation
 - Increase in resource efficiency
 - Promotion of sustainable consumption
 - All of the above
- Q.2 What is the purpose of waste segregation in food processing?
- To reduce waste volume
 - To increase waste disposal costs
 - To facilitate recycling and reuse
 - To improve product quality

Q.3 Which of the following micro-organism is commonly used in fermentation process for by-product utilization?

- a) Bacteria (Lactobacillus)
- b) Yeast (Saccharomyces)
- c) Fungi (Aspergillus)
- d) All of the above

Q.4 Which of the following is a functional property of protein by product like soy-protein isolate?

- a) Provide emulsification
- b) Increase viscosity
- c) Acts as an antioxidant
- d) Enhance colour

Q.5 Which processing technique is commonly used to convert food waste into biofuels?

- a) Fermentation b) Anaerobic digestion
- c) Transesterification d) Pyrolysis

Q.6 How do vegetable peel and skin, as by-product, generally affect food products?

- a) They add a bitter flavour
- b) They contribute antioxidant and dietary fiber
- c) They acts as an emulsifier
- d) They increase the fat content

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SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 The utilization of food by-products can help reduce _____ and promote sustainable food systems.

Q.8 Food by- products can be used as _____ in animal feed, reducing the need for synthetic nutrients.

Q.9 Prebiotic means _____.

Q.10 The non thermal processing methods that use high pressure to inactivate micro-organism and extend the shelf life of by-products are called _____.

Q.11 SFE stands for _____.

Q.12 The hedonic scale is a method used in sensory evaluation to measure the _____ of by products derived foods, with higher scores indicating greater liking.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Give the examples of various food by products and their sources.

Q.14 What are the various types of fermentation occur in by-products preparation.

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